

Advanced Quantitative Text Analysis

Day 5: Fine-Tuning, Causal Inference, and Research Design

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Where We Are

- **Day 1:** representing text
- **Day 2:** concepts, labels, supervised classification
- **Day 3:** unsupervised discovery, topic models, scaling
- **Day 4:** language models, embeddings, transformers, LLMs
- **Day 5:** adaptation, causal inference, and research design

Today

- ① Advanced evaluation: validity, robustness, domain shift
- ② Fine-tuning and model adaptation
- ③ Prompting vs. fine-tuning
- ④ LLM annotation pipelines
- ⑤ Causal inference with text
- ⑥ Text as treatment, outcome, confounder, mediator
- ⑦ Final research design clinic

What Today Is Not

- Not another lecture on precision, recall, and F1.
- Not another lecture on classifiers.
- Not another tour of algorithms.

Today

We move from model performance to research design: when to adapt models, how to validate them, and how to use text in causal research.

The Main Lesson

A text model is not a research design.

A classifier measures variables. A causal design identifies effects. A validation strategy justifies interpretation.

Good text-as-data research connects all three.

Evaluation: We Already Know the Basics

- Day 2 covered:
 - confusion matrices
 - accuracy
 - precision
 - recall
 - F1
 - error analysis
- Today we focus on harder questions:
 - Does the model generalize?
 - Is the measure valid?
 - Is the model stable?
 - Does adaptation help?
 - Can we use text in causal designs?

Performance Is Not the Same as Validity

- A model can predict labels well and still fail as a measure.
- It may learn:
 - topic shortcuts
 - source artifacts
 - annotator bias
 - temporal patterns
 - platform-specific language
 - country or language effects
- The question is not only:

How accurate is the model?

- The question is also:

What exactly is the model measuring?

Advanced Evaluation Questions

- Does the model generalize across domains?
- Does it work across time?
- Does it work across sources?
- Does it work across groups or languages?
- Does it fail systematically for specific cases?
- Are errors substantively consequential?
- Does the measure support the inferential claim?

The Evaluation Target

Target	Question
Prediction	Does the model predict held-out labels?
Measurement	Does the model capture the intended construct?
Generalization	Does it work outside the training setting?
Inference	Does the measure support the substantive claim?
Decision	Is it good enough for the actual use case?

Domain Shift

- Models often assume training and target data come from the same distribution.
- In real research, this is often false.

$$P_{\text{train}}(X, Y) \neq P_{\text{target}}(X, Y)$$

- Examples:
 - train on newspapers, apply to social media
 - train in 2016, apply in 2024
 - train on U.S. politics, apply to Austria
 - train on mainstream outlets, apply to fringe media

Types of Distribution Shift

Shift type	Meaning
Covariate shift	Text changes, label rule stays similar
Label shift	Class proportions change
Concept drift	Meaning of the label changes over time
Domain shift	Genre, platform, or population changes
Annotation shift	Coding standards or annotator behavior changes

Concept Drift

- The same label may change meaning over time.
- Examples:
 - “fake news” before and after 2016
 - “lockdown” before and after COVID-19
 - “AI risk” before and after ChatGPT
 - “populism” across different party systems
- A model trained on old labels may still be accurate on old tasks but invalid for new meanings.

Robustness Checks

- Evaluate by time period.
- Evaluate by source or platform.
- Evaluate by topic.
- Evaluate by country or language.
- Evaluate on out-of-domain data.
- Compare against simpler baselines.
- Inspect whether substantive conclusions change.

Stress Testing Text Models

- Try adversarial or difficult examples.
- Check sensitivity to preprocessing.
- Check sensitivity to prompts.
- Check sensitivity to random seeds.
- Evaluate under class imbalance.
- Check performance on minority cases.
- Read the errors.

Error Analysis Is Substantive Analysis

- False positives can reveal concept ambiguity.
- False negatives can reveal missing linguistic forms.
- Systematic errors can reveal bias.
- Domain-specific errors can reveal poor transportability.
- Error analysis often tells us more than the aggregate score.

What Is Fine-Tuning?

- Start with a pretrained model.
- Add or adapt a task-specific output layer.
- Continue training on labeled data for your task.
- The model updates its parameters to fit the new task.

Simple definition

Fine-tuning adapts a general language model to a specific task or domain.

Pretraining vs. Fine-Tuning

	Pretraining	Fine-tuning
Data	massive unlabeled text	smaller labeled/task data
Objective	language modeling	task performance
Cost	very high	lower/moderate
Purpose	general representation	task adaptation
Example	predict masked/next tokens	classify stance

Adaptation Spectrum

Strategy	What changes?
Zero-shot prompting	only the prompt
Few-shot prompting	prompt plus examples
Retrieval augmentation	prompt plus retrieved context
Feature extraction	downstream model, frozen encoder
Fine-tuning head	classification head
Full fine-tuning	all model parameters
PEFT	small trainable parameter subset

Fine-Tuning Pipeline

- ① Define task and labels.
- ② Create or collect labeled data.
- ③ Split train/validation/test.
- ④ Choose pretrained model.
- ⑤ Tokenize data.
- ⑥ Train task-specific model.
- ⑦ Evaluate on held-out data.
- ⑧ Test robustness and domain transfer.

Fine-Tuning for Classification

- Input text is passed through a pretrained encoder.
- A classification layer maps representation to labels.
- Training updates the classifier and sometimes the base model.

$$\text{text} \rightarrow h_{[\text{CLS}]} \rightarrow \text{softmax}(Wh + b)$$

- Examples:
 - stance classification
 - incivility detection
 - frame classification
 - sentiment analysis

Domain-Adaptive Pretraining

- Continue language-model pretraining on domain-specific unlabeled text.
- Then fine-tune on labeled task data.
- Useful when domain language differs strongly from general corpora.

General LM $\rightarrow$ domain LM $\rightarrow$ task model

- Examples:
 - legal text
 - medical text
 - parliamentary speech
 - social media slang

When Fine-Tuning Helps

- You have a stable task definition.
- You have enough high-quality labeled data.
- Domain-specific language matters.
- Zero-shot or few-shot prompting is unstable.
- You need consistent outputs.
- You need a model that can be run repeatedly and reproducibly.

When Fine-Tuning Is a Bad Idea

- You have very little labeled data.
- Labels are noisy or poorly conceptualized.
- The task is unstable or evolving.
- The base model already performs well enough.
- The cost is not justified.
- You cannot validate out-of-domain performance.

Full Fine-Tuning vs. Parameter-Efficient Fine-Tuning

Full fine-tuning

- update all model parameters
- flexible
- expensive
- higher storage cost
- higher overfitting risk

PEFT

- update small parameter subset
- cheaper
- easier to store/share
- often competitive
- useful with large models

Parameter-Efficient Fine-Tuning

- Instead of updating the whole model, adapt a small part.
- Common approaches:
 - adapters
 - prefix tuning
 - prompt tuning
 - LoRA
 - QLoRA
- Useful when models are large and compute is limited.

LoRA: Intuition

- Large weight matrix W stays frozen.
- Learn a low-rank update:

$$W' = W + BA$$

- A and B are much smaller than W .
- This reduces trainable parameters while adapting behavior.
- The base model remains unchanged.

Fine-Tuning Risks

- Overfitting to small labeled datasets.
- Learning annotation artifacts.
- Catastrophic forgetting.
- Poor out-of-domain performance.
- Leakage between train and test data.
- Expensive hyperparameter tuning.
- False sense of superiority over simpler baselines.

Fine-Tuning Evaluation

- Compare to:
 - majority baseline
 - dictionary baseline
 - classical ML baseline
 - zero-shot prompt
 - few-shot prompt
- Evaluate:
 - held-out data
 - out-of-domain data
 - time split
 - subgroup performance
 - calibration
 - error typology

Prompting vs. Fine-Tuning

	Prompting	Fine-tuning
Data need	low	higher
Cost	low/moderate	higher
Stability	variable	usually better
Reproducibility	harder	easier
Adaptation	instruction-level	parameter-level
Best for	exploration	repeated stable tasks

A Practical Decision Rule

- Start with a simple baseline.
- Try zero-shot or few-shot prompting.
- Validate against human labels.
- If performance is unstable or insufficient:
 - collect more labels
 - fine-tune a smaller model
 - compare against prompting
- Choose the simplest model that supports the research claim.

LLM Annotation Pipeline

- ① Define concept and codebook.
- ② Write prompt with output schema.
- ③ Test on small sample.
- ④ Compare to human labels.
- ⑤ Inspect disagreements.
- ⑥ Revise prompt or label scheme.
- ⑦ Run large-scale annotation.
- ⑧ Validate on held-out human-coded data.

Prompting as Codebook Design

- A prompt defines:
 - concept definition
 - inclusion criteria
 - exclusion criteria
 - examples
 - edge cases
 - output format
- Bad prompts are bad codebooks.
- Good prompts are still not self-validating.

LLM Annotation Failure Modes

- Overly literal interpretation of instructions.
- Overgeneralizing from examples.
- Explaining wrong labels convincingly.
- Sensitivity to prompt wording.
- Sensitivity to ordering of examples.
- Drift across model versions.
- Different behavior across languages or domains.

Reporting LLM-Based Annotation

- Model name and version.
- Date of access.
- Prompt text.
- Sampling settings.
- Output schema.
- Validation sample.
- Human-label comparison.
- Error analysis.
- Robustness to prompt variants.

Human-in-the-Loop Workflows

- LLM labels all documents.
- Humans review uncertain cases.
- Humans review systematic error categories.
- Active learning selects cases for human annotation.
- Final model is validated against human-coded test set.

Goal

Use models to reduce human labor without outsourcing the concept entirely.

Why Causal Inference With Text?

- Many social-scientific questions are causal:
 - Does exposure to negative news reduce trust?
 - Do populist messages increase polarization?
 - Does moral language increase sharing?
 - Do policy frames affect public opinion?
- Text is often central to the treatment, mediator, outcome, or confounder.

Causal Inference Requires a Design

- A classifier does not identify a causal effect.
- A topic model does not identify a causal effect.
- An LLM label does not identify a causal effect.
- Causal identification comes from research design and assumptions.

Key point

The text model measures variables. The causal design identifies effects.

Potential Outcomes

- For each unit i , define:

$$Y_i(1), \quad Y_i(0)$$

- Individual treatment effect:

$$\tau_i = Y_i(1) - Y_i(0)$$

- Average treatment effect:

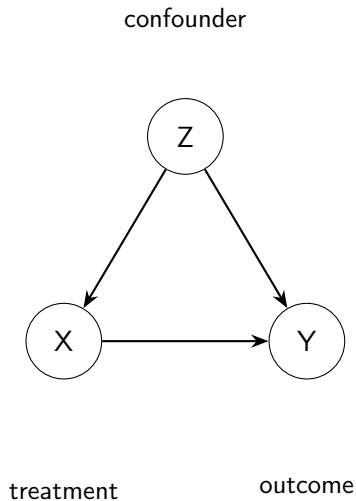
$$ATE = E[Y_i(1) - Y_i(0)]$$

- Problem: we never observe both outcomes for the same unit.

Text Can Play Different Causal Roles

Role	Example
Treatment	exposure to a message frame
Outcome	change in expressed sentiment
Mediator	campaign message explains effect on vote choice
Confounder	topic affects both exposure and outcome
Control variable	text-derived ideology score
Moderator	hostile rhetoric changes treatment effect

DAG: Confounding



Text as Treatment

- Treatment is textual content.
- Examples:
 - exposure to a frame
 - negative vs. positive article
 - populist vs. non-populist message
 - simple vs. complex language
- Best studied with:
 - experiments
 - randomized message exposure
 - controlled text generation

Text as Treatment: Key Challenge

- Text treatments are high-dimensional.
- Changing one feature often changes many others.
- Example:
 - making a message “more populist” may also change negativity, complexity, emotion, and topic.

Question

What exactly is the treatment contrast?

Text as Outcome

- Outcome is measured from text.
- Examples:
 - sentiment in comments
 - toxicity in replies
 - topic attention in media coverage
 - ideological position in speeches
- Main risk:
 - measurement error in the text-derived outcome

Text as Confounder

- Text can measure pre-treatment differences.
- Example:
 - article topic affects both whether a user sees an article and how they react.
- Text-derived controls can help adjustment.
- But bad controls can also introduce bias.

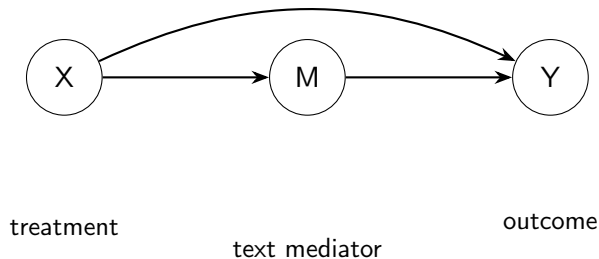
Text as Mediator

- Text can lie on the causal pathway.
- Example:
 - candidate gender affects media framing
 - media framing affects voter evaluation

$$Gender \rightarrow Framing \rightarrow Evaluation$$

- Mediation requires stronger assumptions than total effects.

DAG: Text as Mediator



Text as a Moderator

- Text can change the size or direction of an effect.
- Example:
 - exposure to partisan media may affect trust more strongly when the text uses moral outrage.

$$Y_i = \alpha + \tau T_i + \beta M_i + \gamma(T_i \times M_i) + \epsilon_i$$

- M_i may be a text-derived measure.

Design-Based Thinking

- Define the causal estimand first.
- Draw the DAG.
- Identify treatment, outcome, confounders, mediators.
- Then decide what text model is needed.
- Validate the text-derived variables for that causal role.

Rule

Do not start with the text model. Start with the causal design.

Measurement Error in Text-Derived Variables

- Text models produce imperfect measurements.
- In causal inference, measurement error can bias effect estimates.
- Error can affect:
 - treatment measurement
 - outcome measurement
 - confounder adjustment
 - mediator analysis

Important

Prediction error becomes inferential error.

Outcome Measurement Error

- Suppose the true outcome is Y but we observe text-derived $\tilde{Y}$.
- If error is random, estimates may become noisier.
- If error is systematic, estimates can be biased.

$$\tilde{Y} = Y + e$$

- If e is correlated with treatment, the causal estimate can be distorted.

Confounder Measurement Error

- Text-derived controls are often noisy.
- If a confounder is measured poorly, adjustment is incomplete.
- Example:
 - topic model only partially captures issue context
 - residual topic differences still confound treatment and outcome

Problem

Noisy controls do not fully remove confounding.

Mediator Measurement Error

- Mediation is especially sensitive to measurement error.
- A noisy text mediator can distort indirect effects.
- Example:
 - media framing is measured with a classifier
 - classifier mistakes topic for frame
 - mediation estimate reflects topic, not frame

Sensitivity Analysis

- Ask how conclusions change under plausible measurement error.
- Vary classification thresholds.
- Propagate model uncertainty into downstream estimates.
- Compare alternative text measures.
- Repeat analysis on manually coded subset.
- Report robustness of substantive conclusions.

Two-Stage Text-as-Data Designs

- ① Estimate text model.
 - ② Use predictions as variables in another model.
- Common but risky.
 - Second-stage uncertainty often ignores first-stage measurement error.
 - Best practice:
 - validate first stage
 - propagate uncertainty when possible
 - run sensitivity analyses

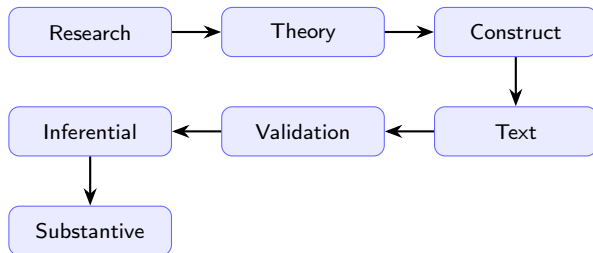
The Social Science Standard

- Not: “I used a powerful model.”
- Not: “The F1 score is high.”
- Not: “The topics look interesting.”

Instead

A good text-as-data study explains how the model, validation, and research design support a substantive claim.

The Full Research Design Chain



Method Choice Matrix

Goal	Likely method
Known concept, transparent rules	dictionary / regex
Known concept, labeled examples	supervised classifier / fine-tuned model
Exploration	topic model / clustering
Semantic dimensions	embeddings / LSS / scaling
Flexible annotation	LLM prompting plus validation
Repeated stable task	fine-tuning
Causal claim	research design first, text model second

Reporting Checklist

- Corpus definition and sampling.
- Preprocessing decisions.
- Model choice and rationale.
- Labels or prompts/codebook.
- Validation strategy.
- Error analysis.
- Robustness checks.
- Limitations.
- How the measure supports the claim.

Reproducibility Checklist

- Code and package versions.
- Random seeds.
- Model versions.
- Prompt versions.
- Data availability constraints.
- API dates and settings.
- Train/validation/test splits.
- Documentation of manual decisions.

Ethics and Data Protection

- Are the texts public or private?
- Are there vulnerable populations?
- Can texts be re-identified?
- Are data sent to external APIs?
- Are labels harmful or stigmatizing?
- Does the model reproduce bias?
- Who is affected by errors?

Final Research Design Clinic

For your own project, answer:

- ① What is the substantive research question?
- ② What is the theoretical construct?
- ③ What role does text play?
- ④ What model will measure it?
- ⑤ How will you validate the measurement?
- ⑥ What is the inferential claim?
- ⑦ What assumptions are required?
- ⑧ What could go wrong?

Project Design Template

Component	Your answer
Research question	
Construct	
Text role	treatment / outcome / confounder / mediator / object
Text model	dictionary / classifier / topic model / embedding / LLM
Validation	
Causal design	experiment / observational / descriptive / predictive
Main risk	

Validation Plan Template

Question	Plan
What is the construct?	
What is the unit of analysis?	
What is the gold standard?	
What error types matter?	
What domain shifts are plausible?	
What robustness checks are needed?	
How will uncertainty be reported?	

Practical Session

- ① Take a project idea.
- ② Identify the role of text.
- ③ Choose a measurement strategy.
- ④ Decide whether prompting, fine-tuning, or classical ML is appropriate.
- ⑤ Draw a DAG if there is a causal claim.
- ⑥ Write a validation plan.
- ⑦ Present the weakest part of the design.

Discussion Questions

- What is the most likely source of measurement error?
- What is the most likely source of causal bias?
- What would a skeptical reviewer attack?
- What simple baseline would be hard to beat?
- What evidence would make you trust the result?

Key Takeaways

- Basic classification metrics are only the beginning.
- Advanced evaluation asks whether models generalize and measure the intended construct.
- Fine-tuning adapts pretrained models to specific tasks and domains.
- Prompting and fine-tuning are alternative adaptation strategies.
- Text can be treatment, outcome, confounder, mediator, or object of study.
- Text models measure variables; causal designs identify effects.
- Measurement error in text-derived variables can become causal bias.
- Strong text-as-data research connects theory, measurement, validation, and design.

The Whole Course in One Sentence

Quantitative text analysis is not about turning text into numbers.

It is about turning text into valid, interpretable, and useful measurements for a research design.

Final Checklist for Any Text-as-Data Project

- ① What is the construct?
- ② What is the unit of analysis?
- ③ What does the model actually output?
- ④ How is the output validated?
- ⑤ What errors matter substantively?
- ⑥ Does the measure generalize?
- ⑦ What claim does the measure support?
- ⑧ What assumptions are required?

Questions?

Questions?

Backup: Calibration

- A model is calibrated if predicted probabilities correspond to observed frequencies.
- If the model predicts 0.8 probability for 100 cases, about 80 should be positive.
- Calibration matters when probabilities are used downstream, not just labels.

Backup: Brier Score

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2$$

- Measures accuracy of probabilistic predictions.
- Lower is better.
- Sensitive to calibration and sharpness.

Backup: Active Learning

- Model selects cases for human annotation.
- Useful strategies:
 - uncertainty sampling
 - disagreement sampling
 - diversity sampling
- Goal: maximize information from limited annotation budget.

Backup: Retrieval-Augmented Generation

- Retrieve relevant documents.
- Insert them into prompt context.
- Generate answer based on retrieved material.
- Useful for document Q&A and structured extraction.

Warning

RAG improves grounding but does not eliminate hallucination or validation needs.

Backup: Bootstrap Uncertainty for Text Measures

- Resample documents.
 - Recompute text-derived measure.
 - Re-estimate downstream model.
 - Inspect variation in results.
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- Useful when analytic uncertainty is hard to derive.